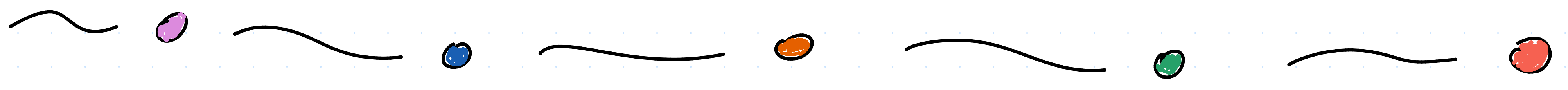


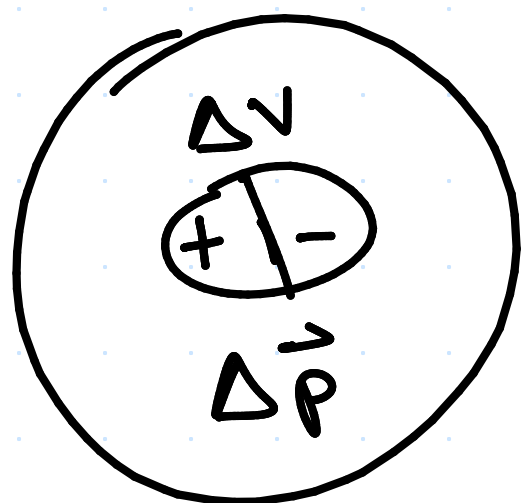
30/09/2025

Clase # 200

Electrodinámica



Unit 2: Electric fields in dielectric media.



$$\Delta \vec{P} = \int_{\Delta V} \vec{r} da$$

$$\vec{P} = \lim_{\Delta V \rightarrow 0} \frac{\Delta \vec{P}}{\Delta V} = \underline{P}(\vec{r})$$

↑
Electric
Polarization

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\vec{P} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dV'$$

$$\nabla \cdot f \vec{F} = f \nabla \cdot \vec{F} + \nabla f \cdot \vec{F}$$

$$\nabla' \cdot \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$

Tenemos

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\int_S \frac{\vec{P} \cdot \hat{n} da}{|\vec{r} - \vec{r}'|} + \int_V - \frac{\nabla' \cdot \vec{P}}{|\vec{r} - \vec{r}'|} dV' \right]$$

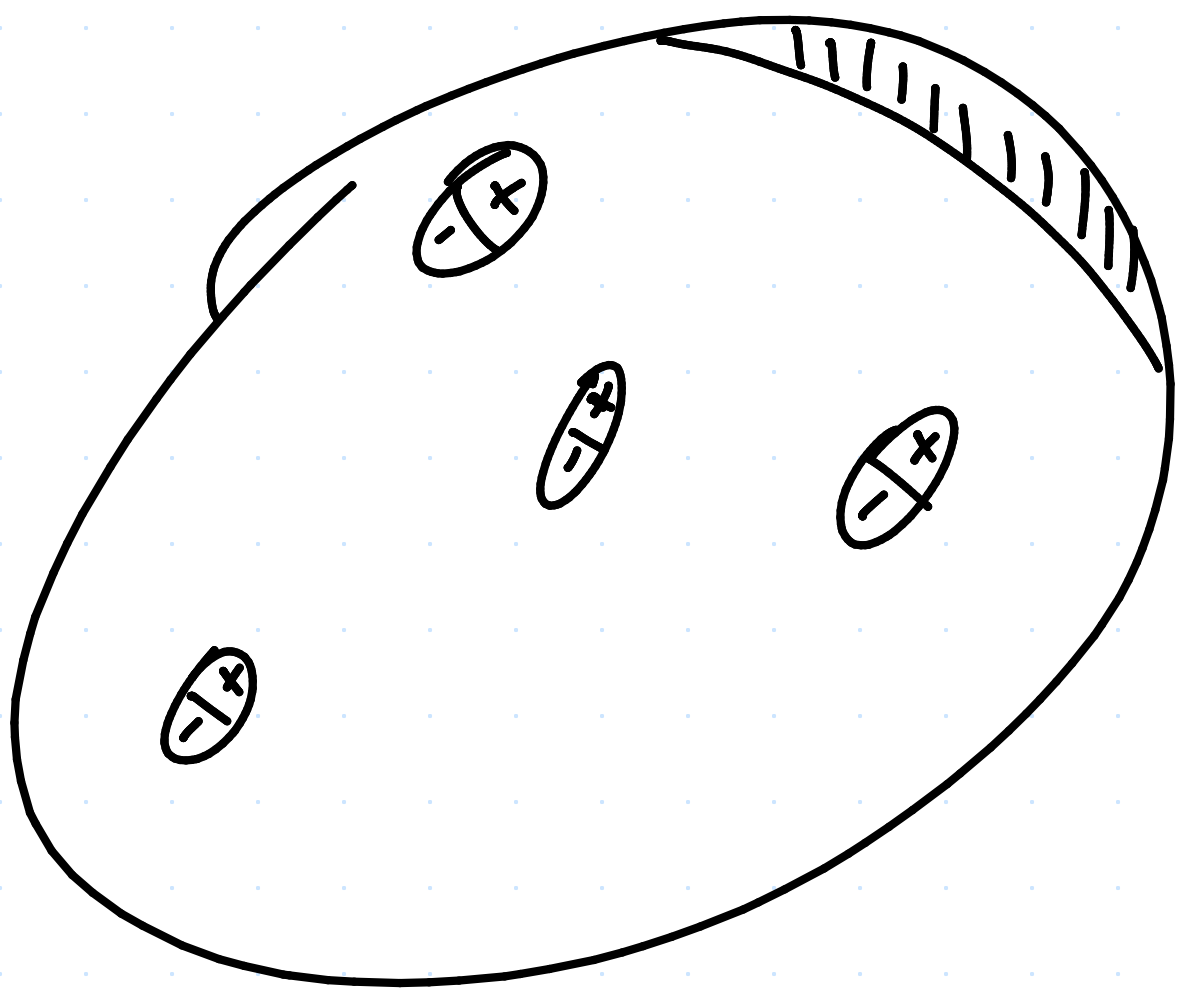
Interpretations

$$\sigma_P = \vec{P} \cdot \hat{n} \quad ; \quad \rho_P = -\nabla \cdot \vec{P}$$

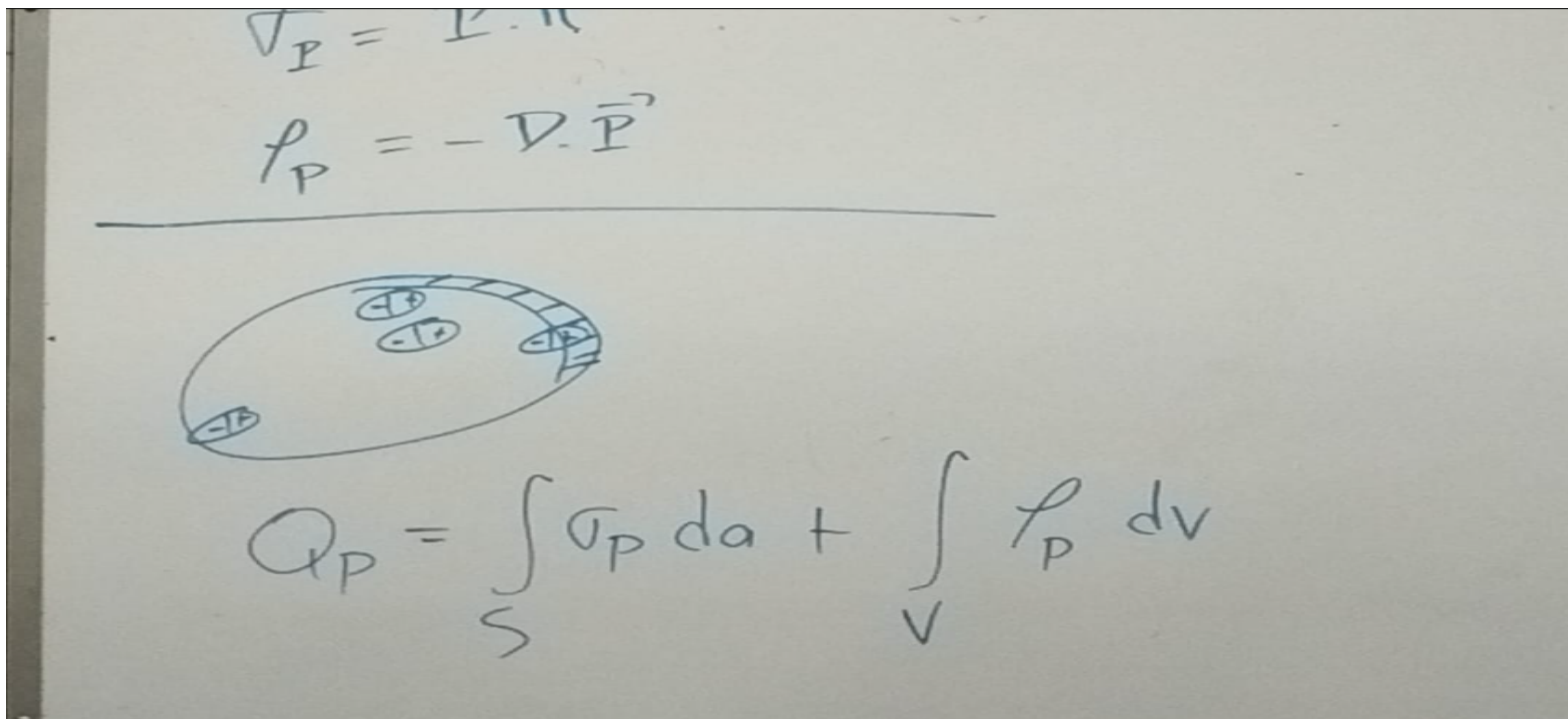
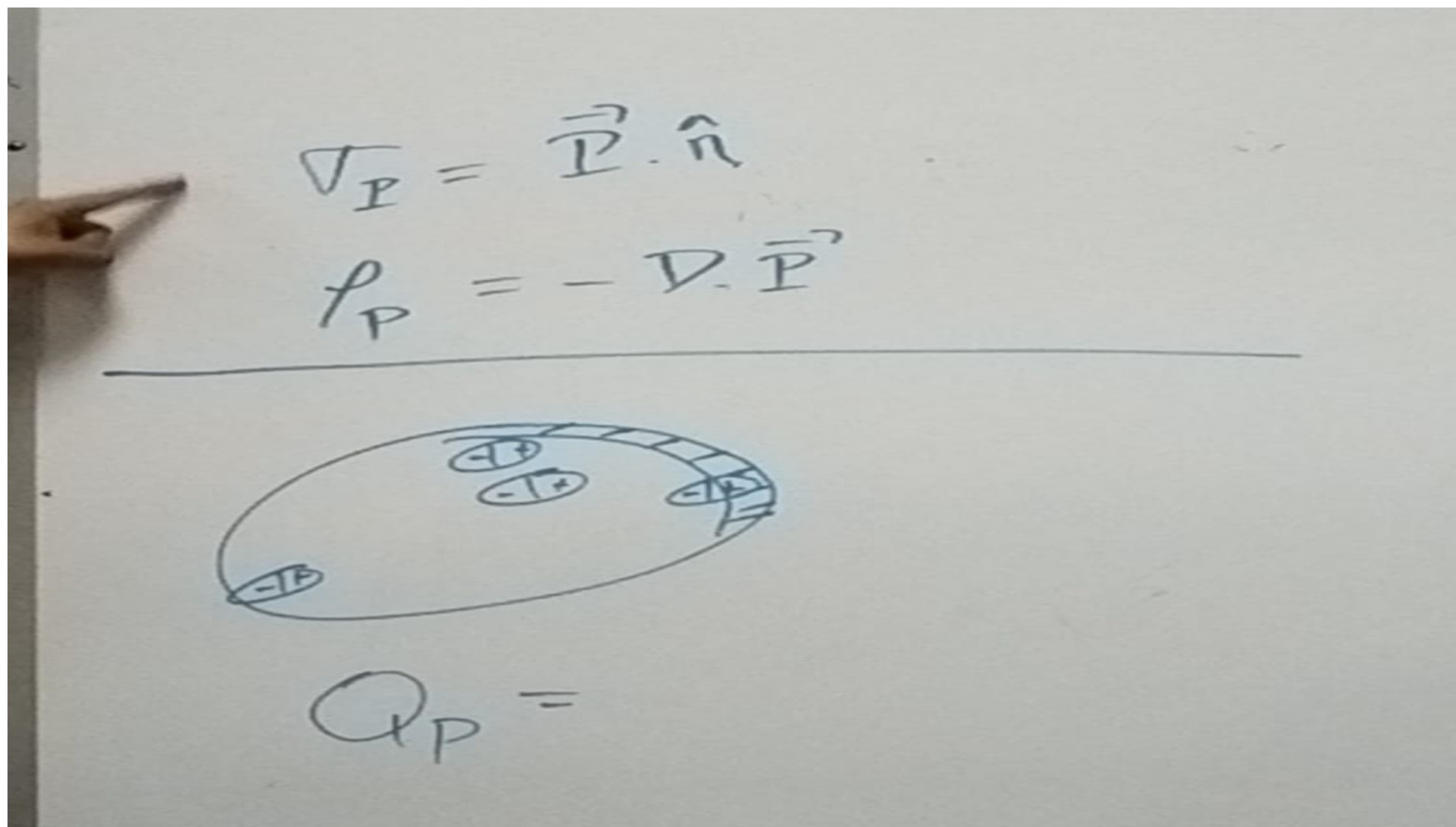
$\uparrow$
Polarización

Reitz - Milford

• Ilustración

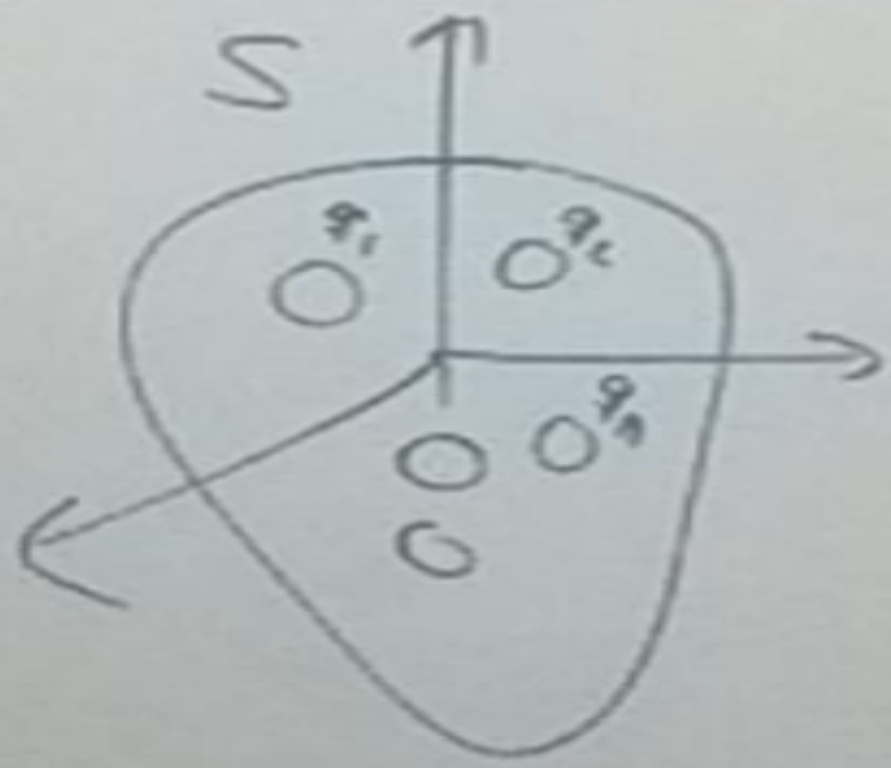


← carga sobre la superficie asociada a la polarización.



Unit 2: Electric fields in dielectric media.

- Gauss' law for dielectrics



$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}')}{|\vec{r} - \vec{r}'|} da + \int_V \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} dv$$

Aplicando el teorema de Gauss se tiene:

$$\int_S \vec{E} \cdot \hat{n} da = Q_p + \sum_{i=1}^n q_i$$

$$= \int_{S_1+S_2+\dots+S_n} \sigma_p da + \int_V \rho_p dv + q/\epsilon_0$$

$$= \int_{S_1+S_2+\dots+S_n} \vec{P} \cdot \hat{n} da + \int_V -\nabla \cdot \vec{P} dv + q/\epsilon_0$$

Aplicando teorema de la divergencia.

$$= \int_{S_1+\dots+S_n} \vec{P} \cdot \hat{n} da - \int_{S+S_1+\dots+S_n} \vec{P} \cdot \hat{n} da + q/\epsilon_0$$

Tenemos

$$\int_S \vec{E} \cdot \hat{n} da = \frac{q}{\epsilon_0} - \int_S \frac{\vec{P} \cdot \hat{n} da}{\epsilon_0}$$

$$\int_s [\epsilon_0 \vec{E} + \vec{P}] \cdot \hat{n} da = q \Rightarrow \boxed{\int_s \vec{D} \cdot \hat{n} da = q}$$

$\vec{D}$ = vector de desplazamiento eléctrico

Definitions:

$$\vec{P} = \chi \vec{E}$$

Electric susceptibility

$$\vec{D} = \epsilon \vec{E}$$

Permittivity

Ejercicio :

Hallar " $\epsilon(x)$ "

Tenemos por:

$$\vec{P} = \chi \vec{E} \quad (1)$$

y por:

$$\vec{D} = \epsilon \vec{E} \quad (2)$$

Despejamos $\vec{E}$ de (1); tenemos

$$\frac{\vec{P}}{\chi} = \vec{E}$$

Tenemos...

$$\vec{D} = \epsilon \left[\frac{\vec{P}}{\chi} \right]$$

Ad !!

"

Queda esto:

$$\underline{\epsilon(x) = \epsilon_0 + \chi}$$

• comentarios

En general: $\epsilon = \epsilon(\vec{E})$; $\epsilon = \epsilon(\omega)$; Dielectric Function

ϵ is a tensor!!!

2-rank tensor

Tenemos → Número dieléctrico del material.

$$\kappa = \frac{\epsilon}{\epsilon_0} =$$

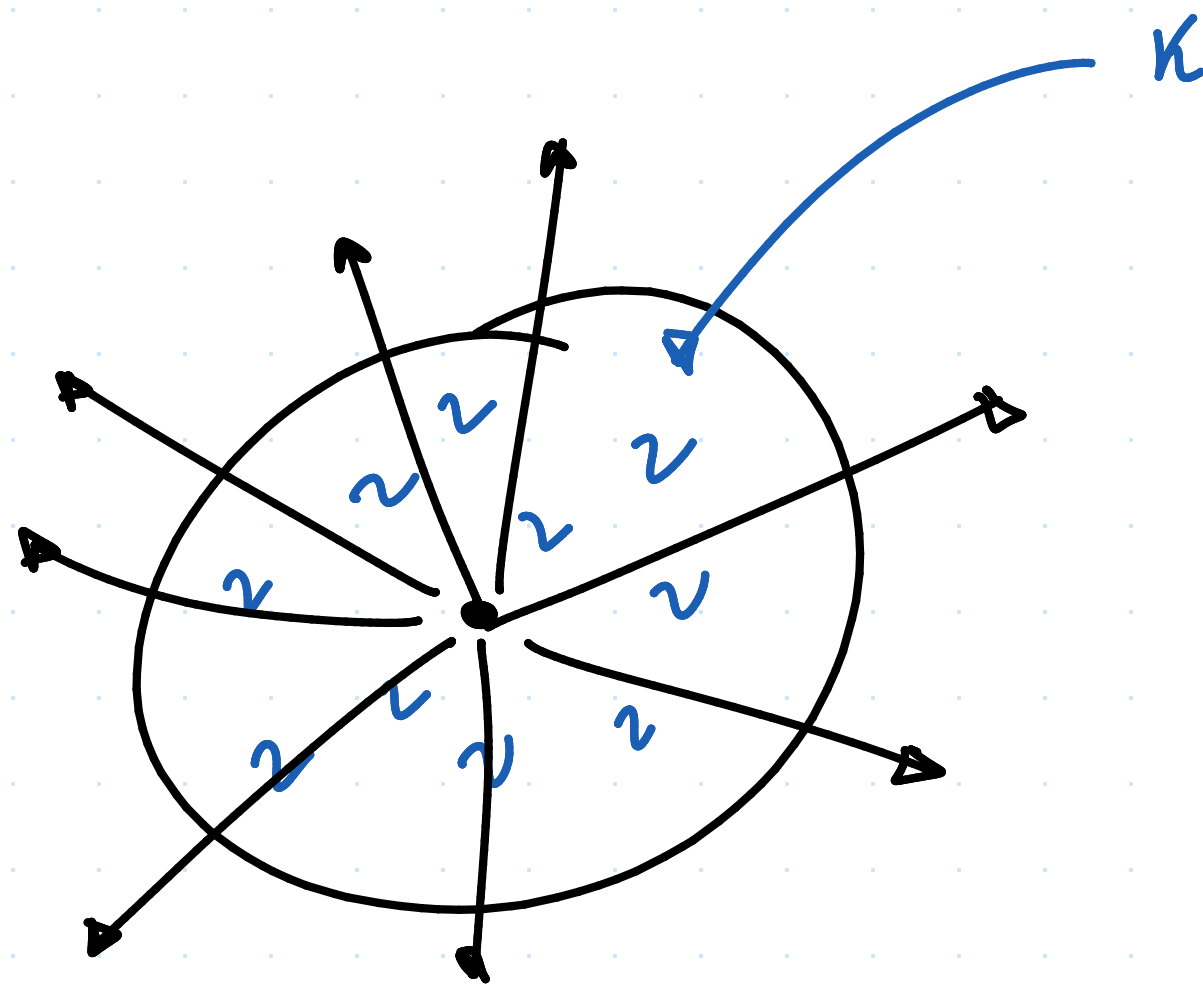
↑
Dielectric
constant

$$\kappa(x) = \frac{\epsilon_0 + \chi}{\epsilon_0} = 1 + \frac{\chi}{\epsilon_0}$$

Exercise: Consider a point charge at the origin and immersed in dielectric media with dielectric constant κ . Find $\vec{E}$ in the dielectric.

1) Find $\vec{D}$:

$$\int_S \vec{D} \cdot \hat{n} da = q$$



$$\hookrightarrow \int_S D da = q \Rightarrow D \int_S da = q$$

$$D(4\pi r^2) = q$$

$$\vec{D} = \epsilon \vec{E}$$

$$D = \frac{q}{4\pi r^2} \hat{r};$$

$$\epsilon \vec{E} = \frac{q}{4\pi r^2} \hat{r}$$

$$\kappa = \frac{\epsilon}{\epsilon_0} \Rightarrow \epsilon_0 \kappa = \epsilon \quad \vec{E} = \frac{q}{4\pi r^2 \epsilon \kappa} \hat{r}$$

Tenemos

$$\vec{D} = \frac{q}{4\pi r^3} \vec{r}$$

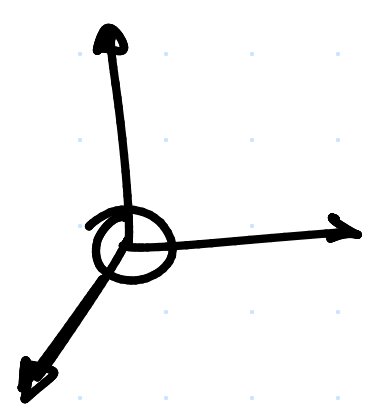
$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} \Rightarrow \hat{r} = \frac{\vec{r}}{r} \rightarrow r \hat{r} = \vec{r}$$

$$\vec{E} = \frac{q}{4\pi \epsilon_0 \kappa r^3} \vec{r}$$

3) Find the polarization; $\vec{P}$

4) Find ρ_P

5)



• idea
• carga puntual en el origen

Tenemos...

$$\int_s [\epsilon_0 \vec{E} + \vec{P}] \cdot \hat{n} da = q \Rightarrow \int_s \vec{D} \cdot \hat{n} da = q$$

$\vec{D}$ = vector de desplazamiento eléctrico

Definitions:

$$\vec{P} = \chi \vec{E}$$

Electric susceptibility

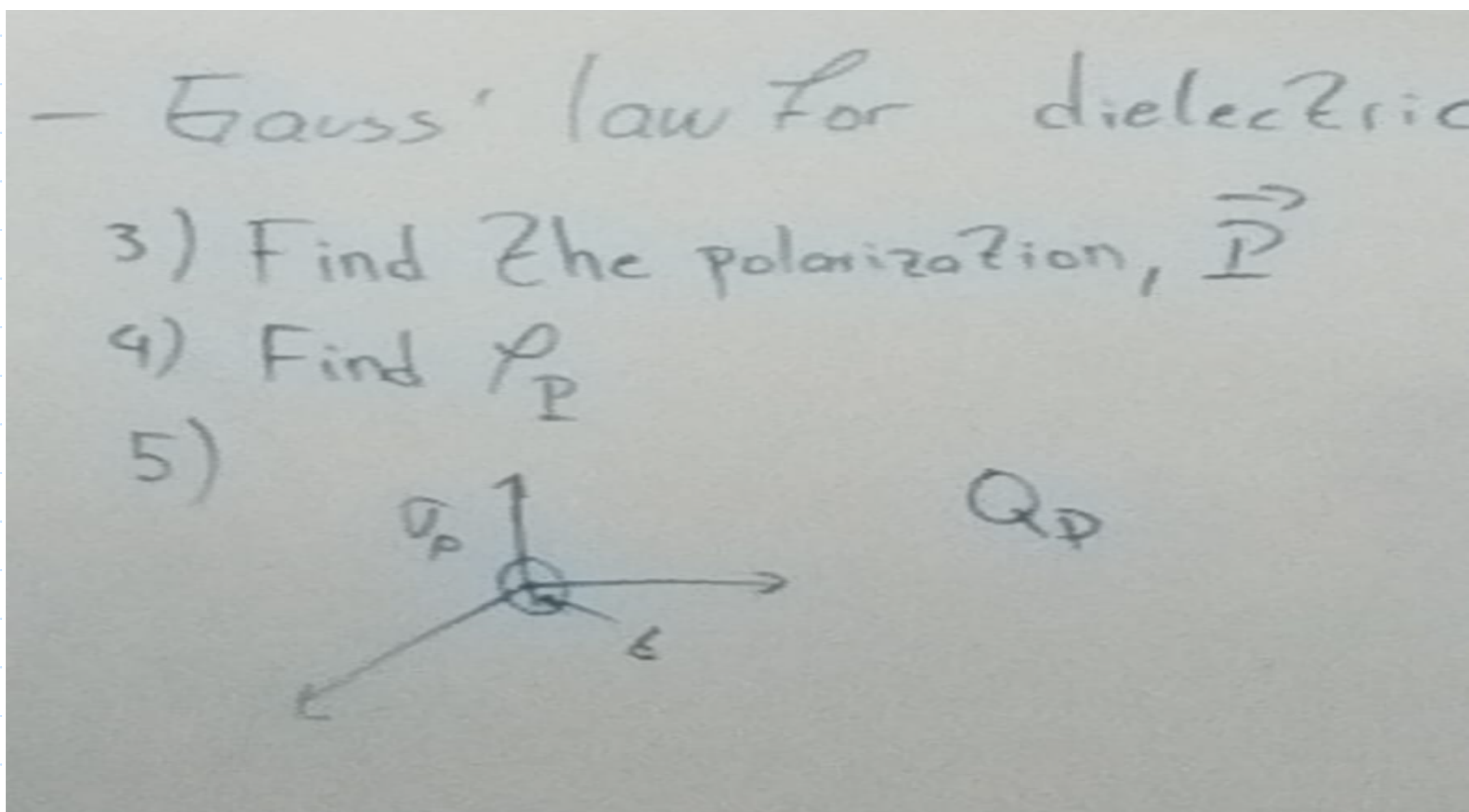
$$\vec{D} = \epsilon \vec{E}$$

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3) Tenemos...

$$\int_s \epsilon_0 \vec{E} \cdot \hat{n} da + \int_s \vec{P} \cdot \hat{n} da = Q_{enc}$$

$$\dots \epsilon_0 \vec{E} (4\pi r^2) + \vec{P} (4\pi r^2) = Q_{enc}$$



• Tarea